Appendix to **IRS 90940**

2nd edition, September 2022

Original

Last update 01/02/2025

Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol

Appendix C

SFERA Implementation Cases

This appendix:

- is updated regularly,
- corresponds to the above date of update.

INTERNATIONAL UNION
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Appendix C - SFERA Implementation Cases

List of implementation cases

Train preparation

- Driver pre-loads the data before the train starts (2 train services) (e.g. loading an S-DAS)

Single driver

- Single driver for single train service without disconnection (proper app quit)
- Single driver for single train service without disconnection (non-proper app quit)
- Single driver for single train service with another person consulting the train journey on another tablet (e.g. instructor)
- Single driver for single train service with lost connection (short)
- Single driver for single train service with lost connection (long)
- Single driver with two successive train services to operate

ATO

- Train service with an ATO, over a “Class B” section of travel

Driver change

- Driver change on single train service (first driver properly disconnects)
- Driver change on single train service (second driver takes over)

Border crossing

- Border crossing
- Driver change at border crossing

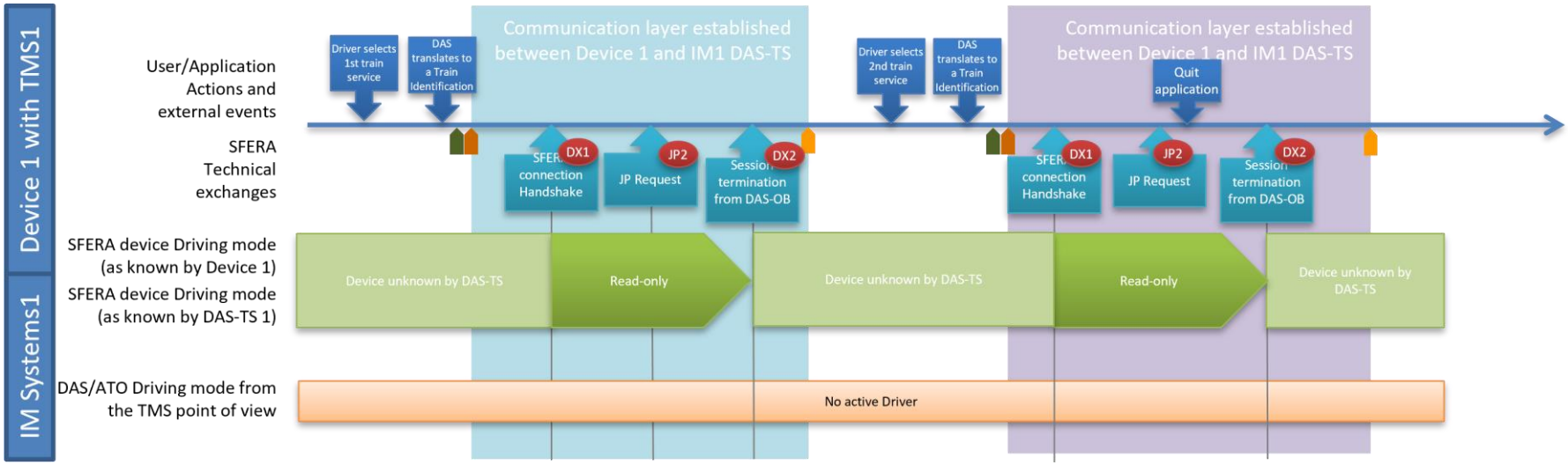
Key

	Graphic Element	Definition
General		External events
General		Systematic first connection sequence that is described in the first implementation cases
Driving mode		Per “Device Driving” mode as seen by the DAS-TS and the device. In cases where the driving mode considered by both systems is different, two arrows are represented in parallel
Driving mode		DAS train driving mode as seen by the TMS. Even if multiple devices are connected to the DAS-TS, the TMS will only consider a single driving mode per train
Use cases		Use case triggered. The implementation cases only reference the first use case triggered by an event, although the given use case can in turn trigger other use cases. For example, a JP2 Journey Profile Request can trigger a JP3 and/or JP4 use case
Use cases		Use case reference
Communication layer		Authorisation procedure that results in an authorisation token for a train movement

	Graphic Element	Definition
Communication layer		Connection on technical level with message broker
Communication layer		Closing of connection with message broker
Communication layer		Communication established between TS and OB

1 Train preparation

1.1 Driver pre-loads the data before the train service starts (two trains on the same IM infrastructure - one JP request per train service)

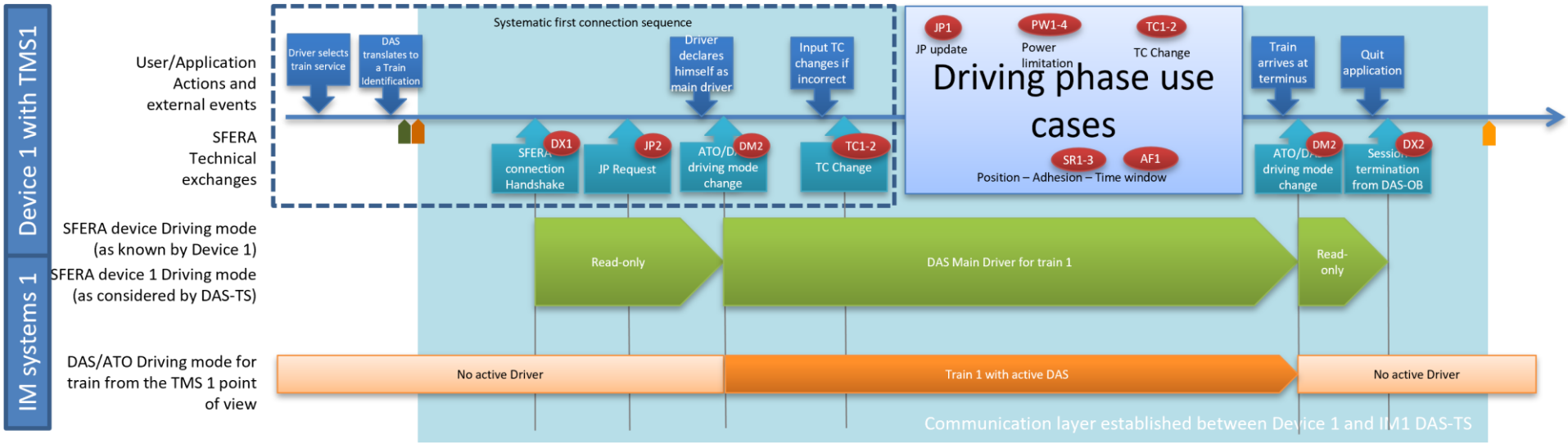


The token references the RU, and the train for which the device is authorised: a SFERA connection between a device and the DAS-TS opens the possibility for the device to access information on an RU's specific train. It is the RU's responsibility to limit the requests of the device to the trains for which the driver is authorised.

Although this implementation case shows the two separate sequences being established and closed sequentially, the protocol allows the device to keep the communication layer open and to start the second JP request sequence with a newly requested token for a second train.

2 Single driver

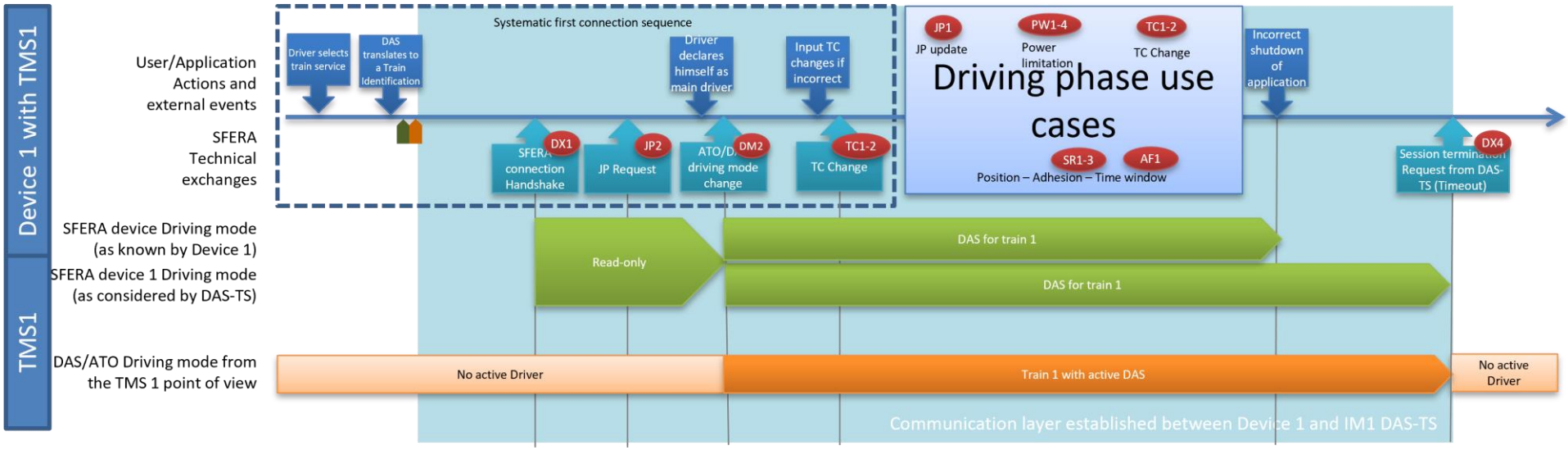
2.1 Single driver for a single train service without disconnection (proper app quit)



Alternatives for the implementation cases:

- When the train arrives at the terminus, the driving mode change could be omitted.
- If DAS is only authorised for the train in question, the device may automatically carry out the disconnection, without going through the “read-only” phase.

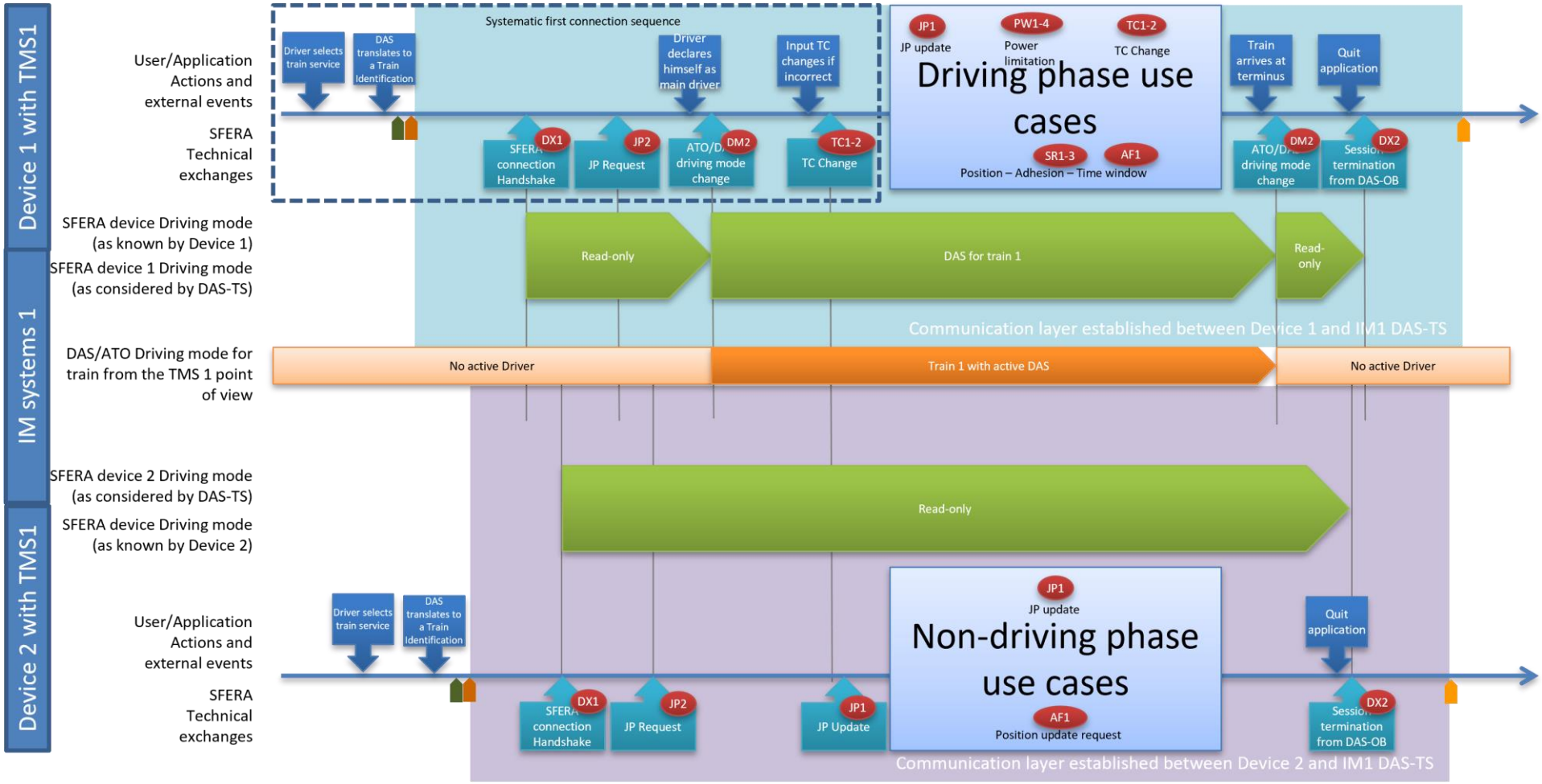
2.2 Single driver for a single train service without disconnection (non-proper app quit)



The connection may also be closed by the IM before the timeout if:

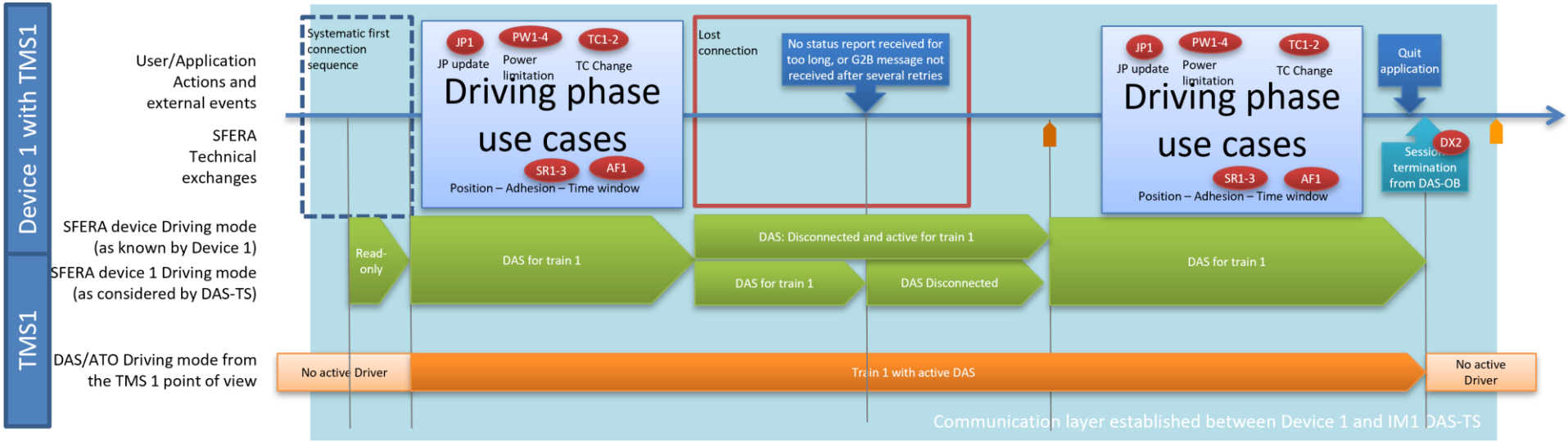
- The authorisation token expires (IM disconnects the device)
- An invalid token is sent, or none at all, with a request/event message

2.3 Single driver for a single train service with another person consulting the train journey on another device (e.g. instructor)



A TC change from the active driver will result in the creation of a new TC in DAS-TS, sent to all devices following that train service (via the JP updates).

2.4 Single driver for a single train service with a loss of connectivity (short)

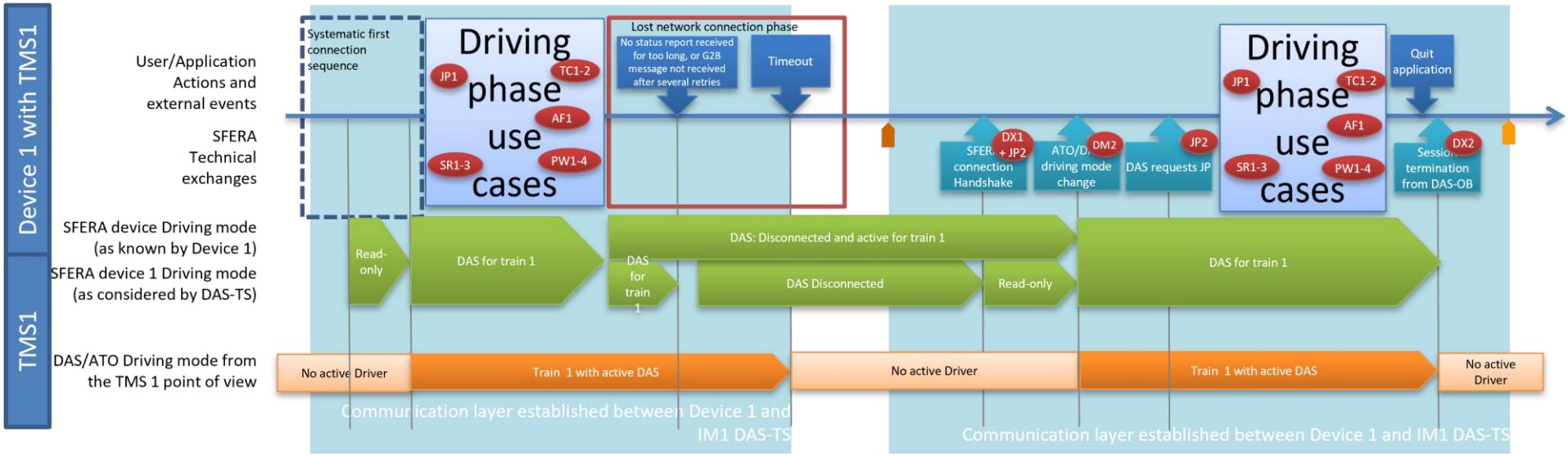


The communication layer does not have a specific requirement concerning immediate detection of a lost connection. Any disconnection is therefore only first detected when the DAS-TS does not receive a status report within the specified time frame.

The communication protocol implements a guaranteed delivery requirement. Upon radio reconnection, depending on the parameters used for MQTT, the device will receive:

- All of the JP updates generated by the TMS while the radio connection was lost
- The last JP update sent

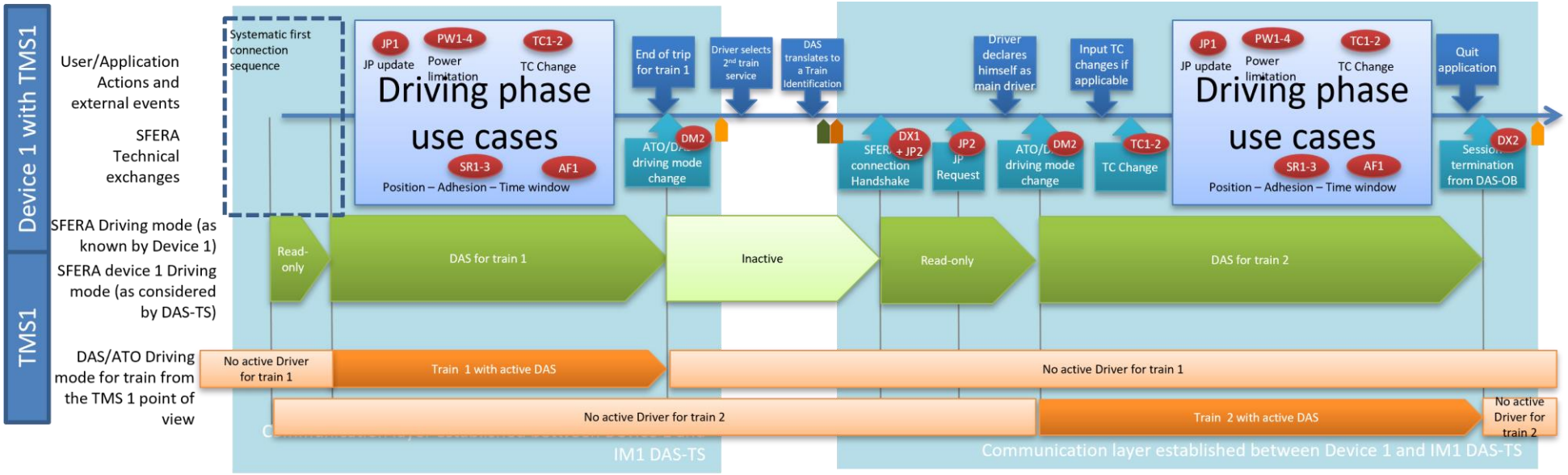
2.5 Single driver for a single train service with a loss of connectivity (long)



Each IM is responsible for determining the time to timeout and the functional actions to be taken upon timeout. They are exchanged during the handshake.

Upon reconnection, the device will automatically reset a connection without repeating the “systematic first connection sequence”. The driver therefore is not required to take action (no train number entry, etc.).

2.6 Single driver with two successive train services to operate

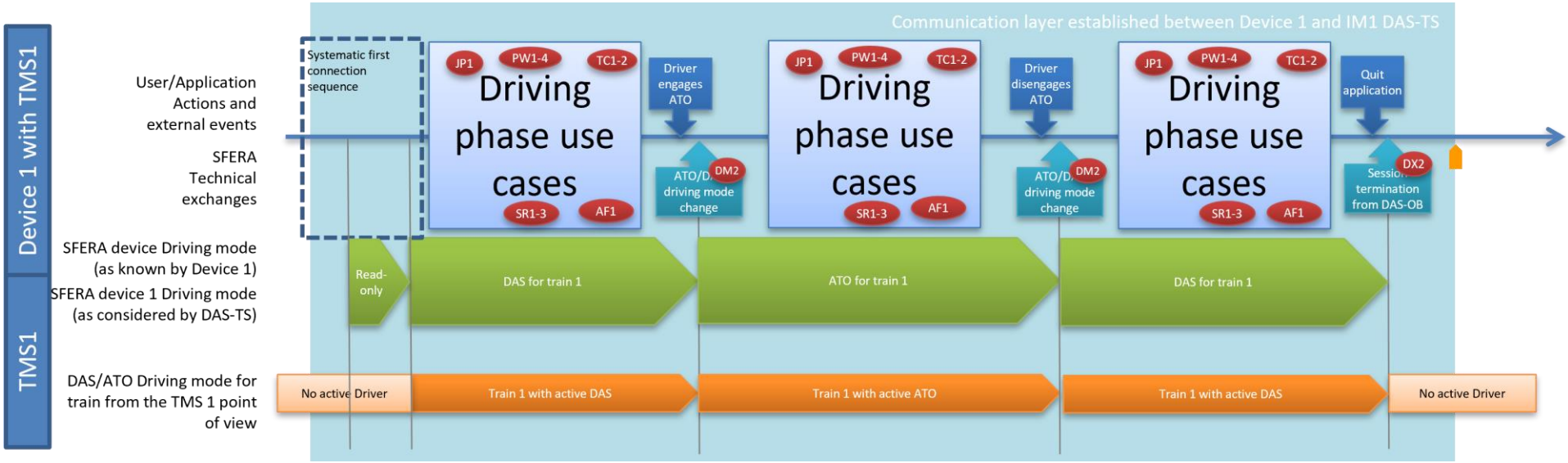


The standard practice for SFERA would be to have a different token for each service.

Therefore, a new token will need to be generated in the connection exchanges for the second train.

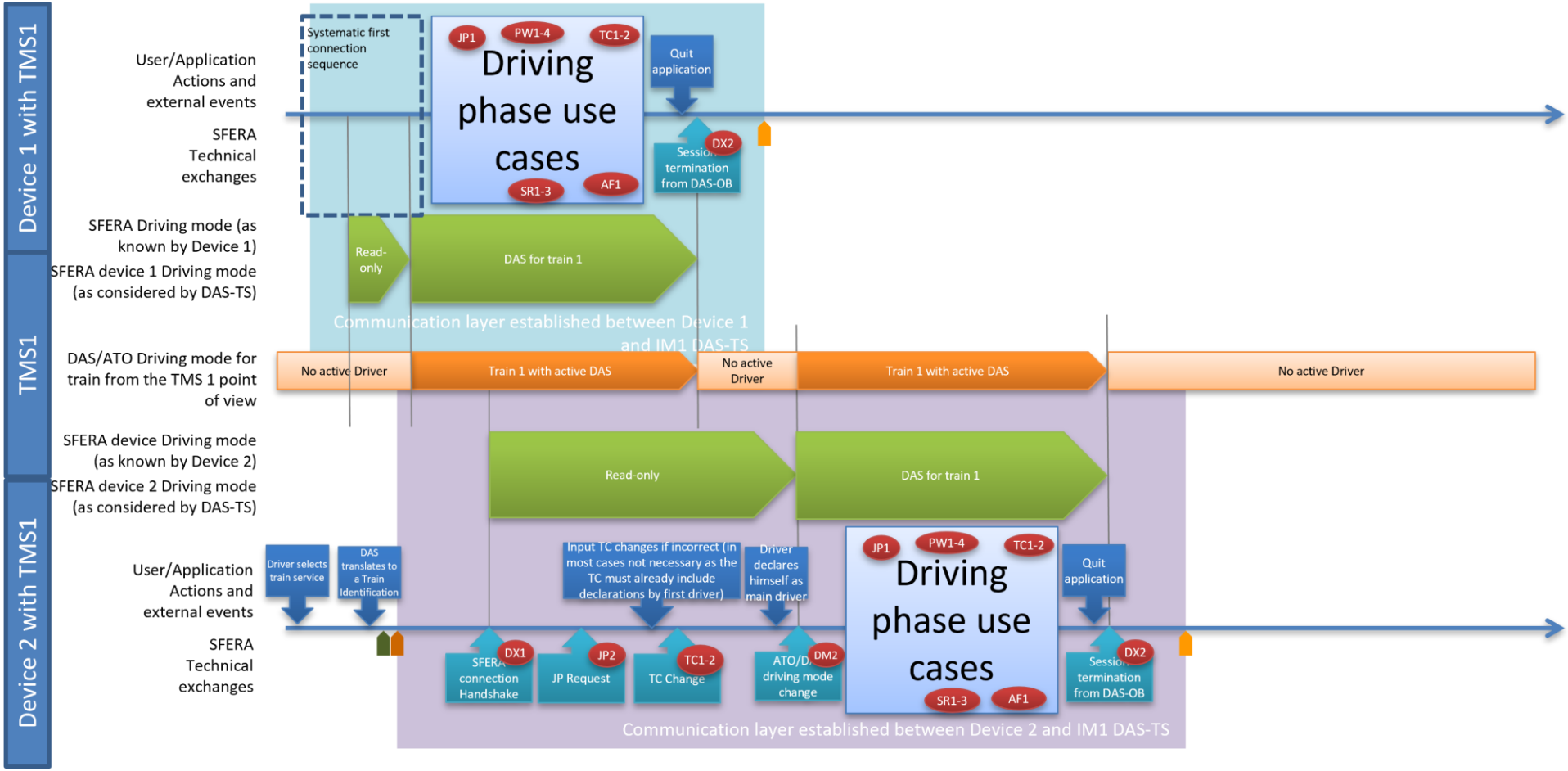
3 Interactions with ATO

3.1 Train service with an ATO, over a “Class B” section of travel (same device)



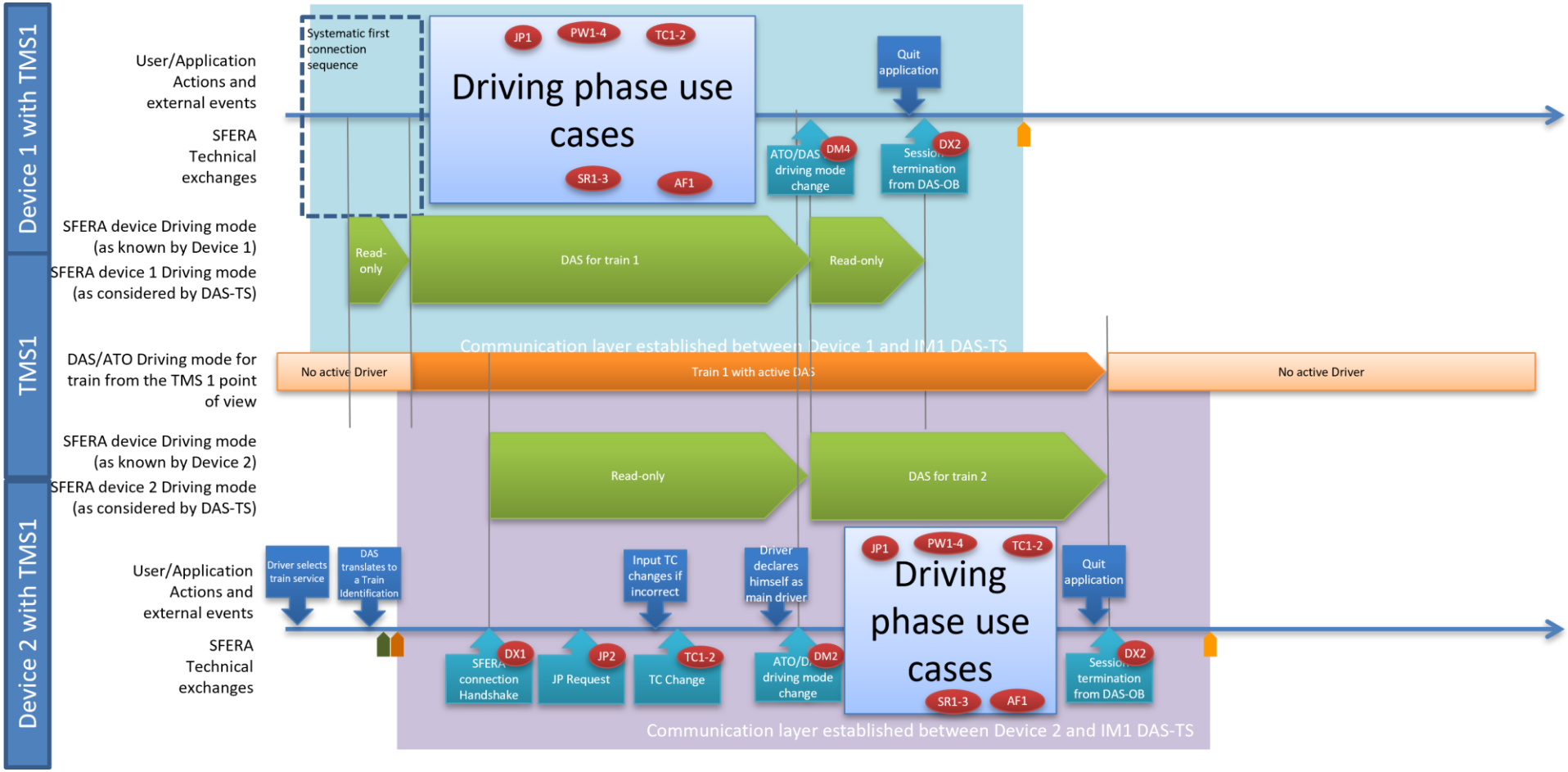
4 Driver changes

4.1 Driver change on a single train service (first driver properly disconnects)



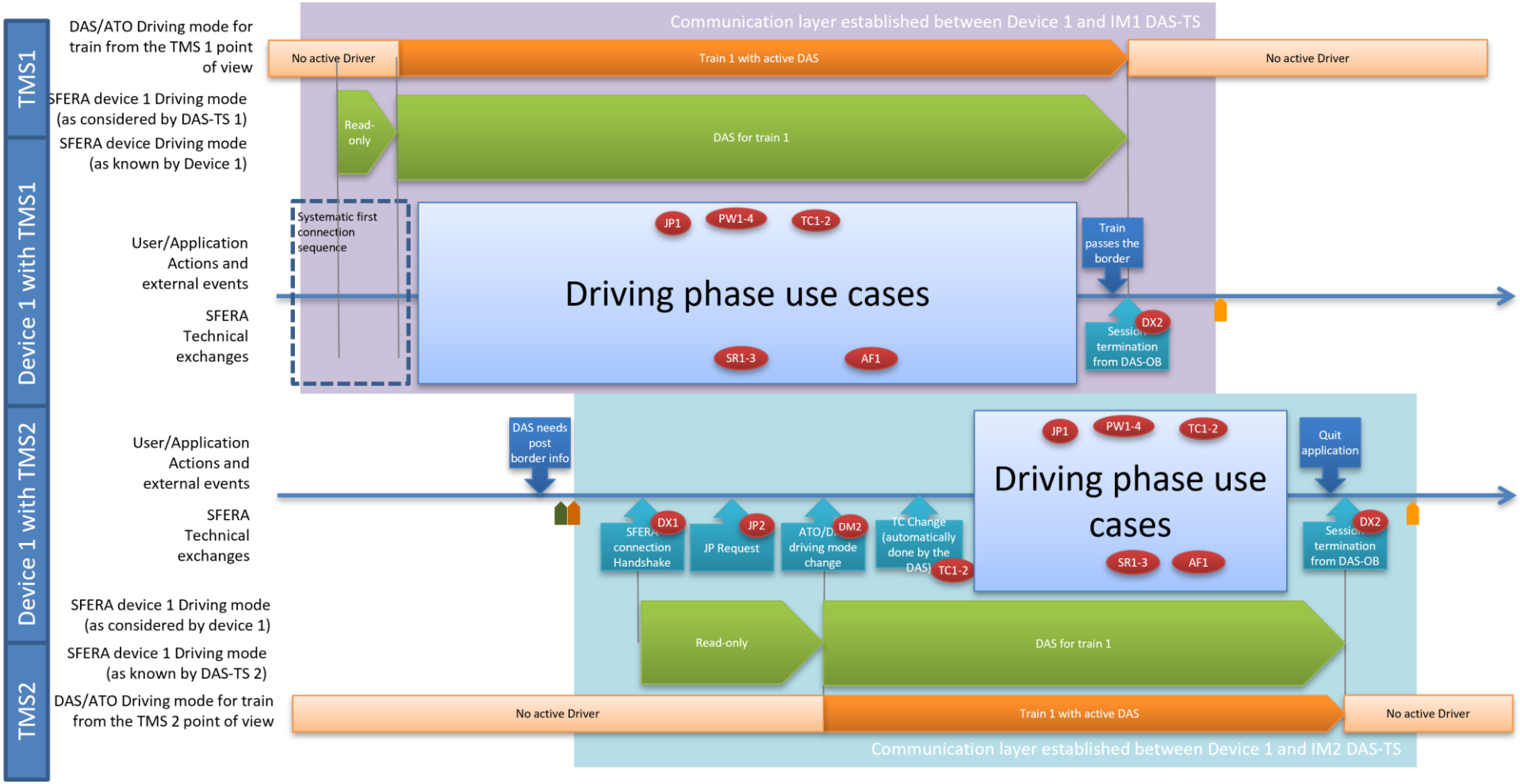
During the phase where Device 2 is in read-only mode and Device 1 is active, Device 2 will receive all JP updates, and will therefore have received all TC changes declared by the driver with Device 1.

4.2 Driver change on a single train service (second driver takes over)



5 Border crossings

5.1 Border crossings (normal)

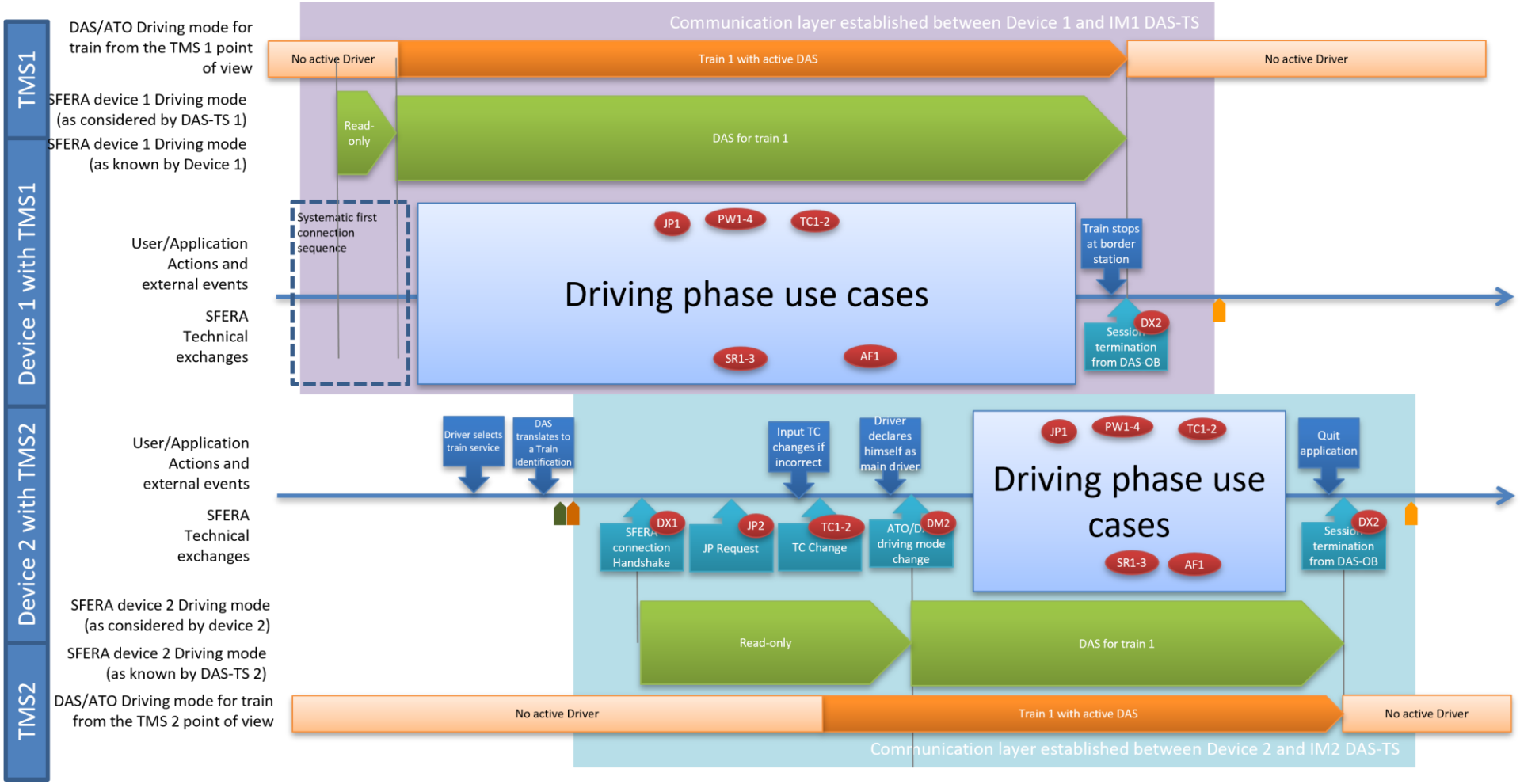


The DAS tries to connect to TMS2 as soon as it considers that it needs data after the border point. This will be before the last timing point before the border, but can also be much earlier (the TMS2 connection could be initiated just after the TMS1 connection).

A border crossing can occur in parallel with an RU change, without a device change: the communication to TMS2 is opened with a different token based on the second RU's key. This implementation applies to a hand-held device or to the integrated DAS of a rolling stock component.

The two RUs have to agree on the conditions of confidentiality under which their respective datasets are to be handled (for example, the possibility for the same application on Device 1 to process data for both RUs at the same time).

5.2 Border crossings with a driver change (person-bound devices)



This implementation case assumes that Device 1 knows that it must not connect to the next IM (information not included in the SFERA protocol). If not, the transition described in the Implementation Cases of section 4 is used.